O-241 10:45 AM Wednesday, October 26, 2022

**THREE-DIMENSIONAL (3D) BIOMIMETIC CULTURE ON POLY(EPSILON CAPROLACTONE) (PCL) PATTERNED SCAFFOLDS: A TISSUE ENGINEERING APPROACH TARGETING EARLY STAGE OF FOLLICULOGENESIS UP TO OOCYTE MATURATION.** Chiara Di Berardino, MS,[1] Alessia Peserico, PhD,[1] Liliana Liverani, PhD, MEng,[2] Giulia Capacchietti, MS,[1] Valentina Russo, PhD, DVM,[1] Nicola Bernabò, PhD, DVM,[1] Aldo Roberto Boccaccini, PhD, MEng,[2] Barbara Barboni, PhD, DVM[1] [1]University of Teramo, Teramo, Italy; [2]Friedrich-Alexander University of Erlangen-Nuremberg, Germany.

OBJECTIVE: A Reproductive Tissue Engineering (REPROTEN) approach has been proposed to sequentially mimic *in vitro* folliculogenesis (*iv*F) and meiotic maturation. To this aim, 3D culture of ovine ovarian follicles are performed on PCL-patterned electrospun scaffolds to recapitulate the native milieu in which the growing follicles span a long-term incubation (14 days) into a biomimetic environment during the transition from preantral (PA) to early antral (EA) stage. This approach leads the enclosed oocyte to end the growth phase before triggering meiotic maturation instead to proceed in their enrollment into a conventional *in vitro* maturation protocol (IVM).

MATERIALS AND METHODS: PA follicles (size diameter: 251±5$\mu$m), mechanically isolated from slaughterhouse lamb ovaries, were individually cultured on previously validated biomimetic PCL-patterned porous scaffolds by placing them on 96-wells plate filled with 100$\mu$L alphaMEM supplemented with 5% Knockout Serum Replacement and 4 IU/mL of equine Chorionic Gonadotropin. At the end of *iv*F, exclusively the EA follicles were divided in 2 different IVM groups: (1) isolated fully-grown oocytes were matured under conventional IVM; (2) intrafollicular maturation were triggered on PCL-patterned scaffolds. The outcomes concern antrum differentiation, the expression of aromatase (CYP19A1) steroidogenic-related gene and meiotic competence.

RESULTS: 3D PCL-patterned scaffolds were able to synchronously sustain follicle and oocyte growth with high viability (88.2%), approximately 72% of antrum cavity differentiation and 1.9-fold change increase of CYP19A1 expression (PA *vs* EA, $p<0.0001$). Furthermore, a greater resumption of meiosis was obtained under the intrafollicular maturation protocol (intrafollicular *vs* conventional IVM: 26.7% *vs* 9.7% Germinal Vesicle Breakdown, $p=0.0012$) even if the incidence of Metaphase II was independent of the IVM protocol adopted (intrafollicular *vs* conventional IVM: 64% *vs* 68% Metaphase II, $p=0.0572$).

CONCLUSIONS: Reproductive biomimetic scaffolds are promising tools to design tissue engineering approaches targeted to innovate *iv*F. The present results confirmed that PCL-patterned electrospun scaffolds can mimic the native ovarian matrix by enabling long-term cultures without affecting 3D-follicle microarchitecture and supporting, at the same time, oogenesis up to the end (maturation phase). The proposed REPROTEN solution provided a multi-organ system in which the cooperation between compartments is preserved as well as the trophic and paracrine mechanism complexity that is essential for controlling this critical phase of the female reproductive cycle (transition from PA to EA and acquisition of meiotic competence).

IMPACT STATEMENT: The development of engineering strategies employed in ovarian science and medicine represents the innovation that Assisted Reproductive Techniques are waiting to solve the increasing demand of female fertility through an improvement of *iv*F to functional recover the large pool of early-stage follicles on individual basis.

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**REPLICATION OF MAMMALIAN GAMETES OF BOTH SEXES IN A MOUSE MODEL.** Philip Xie, B.S., Zev Rosenwaks, M.D., Gianpiero D. Palermo, M.D., Ph.D. The Ronald O. Perelman and Claudia Cohen Center for Reproductive Medicine, Weill Cornell Medicine, New York, NY.

OBJECTIVE: To assess the full preimplantation embryo developmental competence of identical replicates of both male and female gametes.

MATERIALS AND METHODS: Copies of oocytes were initiated by parthenogenetic activation of metaphase II oocytes from B6D2F1 mice by ionomycin. The male counterparts were generated by injecting B6D2F1 spermatozoa into enucleated ooplasts. Haploid constructs of both sexes were cultured to blastocysts, and haploid stem cell lines (haESCs) were derived.

In a different cohort, haploid blastomeres at the 8-cell stage were used as identical copies of female gametes by nuclear transfer into recipient ooplasts to generate putative artificial oocytes. These artificial oocytes were inseminated with wildtype mouse spermatozoa to generate conceptuses. Similarly, 8-cell haploid androgenetic blastomeres were fused with an intact oocyte as a male genome donor. Both types of constructs were cultured in a time-lapse system and compared with control ICSI conceptuses. Ploidy was assessed on the haploid blastomeres and the derived stem cell lines.

RESULTS: Parthenogenetic activation was performed on 129 oocytes and generated 119 (92.2%) constructs with a female pronucleus. Androgenetic counterparts were generated from piezo-ICSI on 375 ooplasts and yielded 317 (84.5%) constructs with a male pronucleus. The pronuclear decondensation for both haploid experimental constructs was comparable to the controls at 90.7%. However, compared to the blastocyst development rate of the controls at 80.8%, haploid female and male embryos yielded blastocysts at rates of 27.8% and 11.2%, respectively ($P<0.00001$). A total of 5 parthenogenetic and 3 androgenetic haploid stem cell lines were derived, and 80% of cells karyotyped maintained haploidy.

In a different cohort, haploid embryos of parthenogenetic and androgenetic origins were cultured up to the 8-cell stage at a respective rate of 76.4% and 59.3%. A total of 105 putative oocytes generated from 15 replicated oocytes yielded a fertilization rate of 88.6% (93/105) and an eventual blastocyst rate of 76.6% (71/93). For the male counterparts, a total of 145 biparental constructs were generated from 21 androgenetic embryos and 88 (60.7%) blastocysts were achieved. All reconstructed embryos retained morphokinetics that were similar to control conceptuses.

CONCLUSIONS: Haploid embryos of both sexes can achieve full preimplantation development and generate haploid stem cell lines with retained haploidy. In our model, a single haploid embryo stemmed from a single gamete can yield 4–5 embryos with retained preimplantation development and identical haplotype.

IMPACT STATEMENT: Male and female genomes from a respective spermatozoon or oocyte can be replicated using our proposed technique; resulting embryonic cells can be used for reproductive applications. Once our technique is optimized in humans, it can be used to investigate gamete heterozygosity for inherited disorders.

SUPPORT: None.

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**BASEMENT MEMBRANE HETEROGENEITY INFORMS AXIS DETERMINATION IN PREGASTULATING HUMAN EMBRYO.** Dong-Yuan Chen, Ph. D,[1] Shiny Titus, PhD,[2] Bailey A. T. Weatherbee, B.S.,[3] Rachel S. Mandelbaum, MD,[4] Richard T. Scott, Jr., M.D.,[5] Emre Seli, MD,[5] Magdalena Zernicka-Goetz, Ph. D[1] [1]California Institute of Technology, Pasadena, CA; [2]Foundation for Embryonic Competence, Basking Ridge, NJ; [3]Department of Physiology, Development and Neuroscience, Mammalian Embryo and Stem Cell Group, University of Cambridge, Cambridge, United Kingdom; [4]University of Southern California, Los Angeles, CA; [5]IVI RMA New Jersey, Basking Ridge, NJ.

OBJECTIVE: We aim to elucidate how extracellular cues contribute to the establishment of the anterior-posterior (AP) axis in the human embryo, which is critical during early pregnancy.

MATERIALS AND METHODS: We used an improved *in vitro* culture system to culture vitrified human blastocyst-stage embryos (n=60) to post-implantation stages (7 to 11 days post-fertilization (dpf)). To visualize the extracellular architecture, embryos were immunostained for LAMININ, a component of the basement membrane (BM), and subjected to 3D super-resolution imaging. Additionally, cell identities were assessed with immunostaining for OCT4 (which marks the epiblast – the precursor of the fetus and amnion), GATA6 (which marks the extra-embryonic hypoblast – the precursor of the yolk sac), and CER1 (which marks the embryo's putative anterior side). Embryos that developed proper tissue architecture and contained the proper cell types were analyzed. We utilized machine-learning-based imaging processing algorithms to quantitate properties of the BM and to correlate these properties with pattern formation. To functionally test how the BM shapes embryonic patterning, we spatially manipulated the BM in mouse embryos at pre-gastrulating stages (n=90) and performed time-lapse live imaging with these embryos *ex utero*. Welch's t-tests were used for statistical analyses.

RESULTS: Single-cell RNA sequencing revealed that genes for components of the BM are highly upregulated in the hypoblast, and 3D super-resolution imaging confirmed that there is a LAMININ+ BM that lies between the OCT4+ epiblast and the GATA6+ hypoblast. This BM is present at 7 dpf and at 9 dpf, indicating that it is formed during the peri- to post-implantation transition. Strikingly, the BM exhibits spatially heterogeneous patterns during post-implantation stages. This heterogeneity occurs before signs of AP axis formation becomes evident. By manipulating the BM in the mouse pre-gastrulating embryos, we demonstrated that this heterogeneity in the BM shapes the collective cell migration event that establishes the AP axis. Following disruption of the BM, aberrant migration of anterior-specifying cells leads to mislocalization of posterior features such as the primitive streak. Using 3D quantitative imaging analyses to define cell behaviors, we showed that the BM informs collective cell migration by tuning tissue fluidity during morphogenesis.

CONCLUSIONS: Our results showed that the BM exhibits spatial heterogeneity during peri- to post-implantation embryogenesis, and that this heterogeneity informs the formation of the AP axis. These results reveal a conserved mechanism in which the extracellular mechanical anisotropy shapes body axis determination in pre-gastrulating embryogenesis.

IMPACT STATEMENT: We discovered a novel mechanism by which extracellular cues regulate pattern formation at critical stages in early pregnancy. Our findings identify morphogenetic indicators that could be applied when evaluating the developmental potential of an embryo.

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**EVALUATION OF CRISPR/Cas9-INDUCED DNA BREAK REPAIR OUTCOMES IN MOUSE AND HUMAN EMBRYOS.** Bieke Bekaert, M. Sc.,[1] Annekatrien Boel, PhD,[2] Lisa De Witte, M. Sc.,[1] Gwenny Cosemans, M. Sc.,[1] Susana M. Chuva De Sousa Lopes, Ph.D.,[3] Björn Menten, Prof. PhD,[1] Björn Heindryckx, Prof. Ph.D.[1] [1]Ghent University; [2]Ghent University, Gent, Belgium; [3]Leiden University Medical Center, Leiden, Netherlands.

OBJECTIVE: Comparison of the DNA repair outcomes following CRISPR/Cas9 gene editing in mouse and human, aiming for the correction of a male infertility-causing *Plcz1*/*PLCZ1* mutation.

MATERIALS AND METHODS: For mouse, we employed a previously described homozygous *Plcz1* knock-out model ($Plcz1^{\Delta 22bp}$). For human, sperm of a patient experiencing fertilization failure after routine ICSI, harboring a heterozygous *PLCZ1* mutation (c.136-1G>C), was utilized. For both mouse and human, a CRISPR/Cas9 guideRNA molecule was designed that specifically recognized the respective mutant allele. The guideRNA-Cas9 complex and a repair template harboring the wild type *Plcz1*/*PLCZ1* sequence was injected into the oocyte. To overcome fertilization failure, AOA was applied during ICSI. At the eight-cell stage, a number of embryos were dissociated in individual blastomeres. For the remaining embryos, after a culture period of maximal 4 or 6 days in mouse and human, respectively, the DNA was extracted and analyzed with targeted and genome-wide sequencing techniques.

RESULTS: In both mouse and human, CRISPR/Cas9 editing did not affect embryonic development. In mouse, assessing the target region, 24% (5/21) of the obtained blastocysts still contained the unedited $Plcz1^{\Delta 22bp}$ allele, 24% (5/21) harboured additional indel mutations and 52% (11/21) contained the wild type sequence. Template use could be identified in (27%) 3/11 of the embryos. In human, 87 embryos were assessed. Due the heterozygous nature of the *PLCZ1* mutation, approximately 50% (44/87) of the embryos originated from mutant sperm. At the target region, 14% (6/44) showed the untargeted mutant allele, 50% (22/44) additional mutagenesis and 36% (16/44) the wild-type allele. In none of these alleles, repair template use was observed. This suggests that either the maternal wild type allele has been used as a repair template (interhomologue homologous recombination (IH-HR)), which has been indeed observed in our human dataset, or that only the maternal allele has been detected ((partial) loss of the paternal targeted chromosome). This is currently further investigated in both mouse and human by (whole genome-wide) single nucleotide polymorphism analysis.

Deep sequencing of whole mouse and human embryos has also pointed to the occurrence of mosaicism, which is the presence of two or more genetically different sets of cells. This was more reliably investigated by single cell analysis. In 6/7 eight-cell stage human embryos originating from mutant sperm, mosaicism was observed. Assessment in mouse is still ongoing.

CONCLUSIONS: The use of a mouse model to reliably study repair mechanisms as they would occur in all types of mammalian embryos might be hampered by the difference in repair template use between mouse and human.

IMPACT STATEMENT: Our findings demonstrate caution to use CRISPR/Cas9 in the human germ line. Firstly, they seem to contradict other reports that show predominant lack of mosaicism. Secondly, the identification of IH-HR events in our dataset contributes to the currently ongoing discussion of which DNA repair mechanisms occur in CRISPR/Cas9-targeted mammalian embryos.

SUPPORT: We want to acknowledge the financial support by FWO-Vlaanderen (Flemish fund for scientific research) for B.B. (11C2821N) and A.B. (1298722N). We also want to thank Ferring Pharmaceuticals (Aalst, Belgium) for their unrestricted educational grant.

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**DE NOVO GAMETES GENERATED IN A NOVEL THREE-DIMENSIONAL CULTURE SYSTEM TO CREATE EMBRYOS CAPABLE OF FULL PREIMPLANTATION DEVELOPMENT.** Mary McKnight, B.S.,[1] Philip Xie, B.S.,[2] Zev Rosenwaks, M.D.,[2] Gianpiero D. Palermo, M.D., Ph.D.[2] [1]Weill Cornell Medicine, New York, NY; [2]The Ronald O. Perelman and Claudia Cohen Center for Reproductive Medicine, Weill Cornell Medicine, New York, NY.

OBJECTIVE: To obtain *de novo* male gametes capable of supporting full preimplantation development utilizing a novel three-dimensional (3D) culture system.

MATERIALS AND METHODS: Male mouse embryonic stem cells (mESCs) were cultured on a gelatin-coated 6-well plate with fibroblasts in monolayer and later spherified by resuspension in a sodium alginate solution and dropped in a calcium chloride solution. Spheres of mESCs were submerged in specifically designed conditioned medium containing Activin A, bFGF, and KSR for 3 days to encourage differentiation into epiblast-like cells, followed by medium containing BMP4, LIF, SCF, EGF, and retinoic acid for up to 36 days to encourage differentiation into male germ-like cells. Cells were assessed for germ cell differentiation biomarkers and injected into oocytes on days 15, 22, 29, and 36, as normal spermatogenesis occurs in 30 days. Assessed markers included DAZL (spermatogonium), VASA (primary spermatocyte), BOULE (post-meiotic secondary spermatocyte), and acrosin (spermatid). Differentiated cells with diameters of approximately 7 mm were injected into oocytes and activated by calcium ionophore. Fertilized conceptuses were monitored in a time-lapse incubator comparing to a control cohort of embryos generated by piezo-ICSI using mature epididymal spermatozoa.

RESULTS: Advancement into *in vitro* spermatogenesis was exhibited on day 3 by the expression of DAZL in 20% and VASA in 15% of spherified cells. On day 10, DAZL and VASA were reassessed and increased to 45% and 18% expression, respectively; post-meiotic biomarkers BOULE and acrosin were manifested in a scant number of cells at 2% and 1%, respectively. On day 15, VASA expression plateaued at 17%, BOULE expression peaked at 10%, and acrosin reached 5%. At the fourth assessment on day 22, expression of VASA increased slightly to 19%, BOULE decreased to 8%, and acrosin peaked at 7%. On day 29, VASA expression peaked at 20%, BOULE expression dropped to 2%, and acrosin expression remained stable at 7%. On day 36, VASA expression remained at 13%, and BOULE and acrosin expression fell drastically to 0% and 1%, respectively.

The control cohort attained 89.2% fertilization and 77.8% blastocyst rates. Generated putative post-meiotic cells on days 15, 22, 29, and 36, equivalent to BOULE- and VASA-expressing cells of approximately 7 mm in diameter, were injected into oocytes. Following chemical activation by calcium ionophore, fertilization was achieved at rates of 35.0%, 61.1%, 81.8%, and 75.0%, correspondingly. Respective blastocyst rates according to the four time intervals were 5.0%, 16.7%, 36.4%, and 8.3%.

CONCLUSIONS: Our novel 3D differentiation model can generate competent gametes to support full pre-implantation development and can obviate the need for final germ cell maturation through allo-/xenogeneic transplantation.

IMPACT STATEMENT: If the results are reproducible and the ability to obtain healthy offspring is confirmed, this method may represent a feasible model for achieving neogametogensis in humans.